CSCI 5541 NLP HW6

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1 Introduction

The goal is to improve your knowledge of signal processing and Fourier Transforms by working with sound files directly. I use Google Text-to-Speech Suite (gTTS) generates speech from the specified English question and saves it as a temporary MP3 file. It then calls the ffmpeg tool to convert the MP3 into a standard WAV audio file (speech.wav) with an 8000 Hz sampling rate, mono, and 16-bit format for subsequent audio feature analysis. And the Audio Info is following.

- **Content** "Are you going to the party tonight?"
- **Sample Rate:** 8000 Hz
- **Channels:** 1 (Mono)
- **Total Samples:** 20736
- **Audio Length:** 2.59 seconds

2 Spectrogram

To analyze the 8000 Hz audio signal, we first split the continuous sound into short, overlapping segments. Each segment uses a 25 ms window and moves forward by a 10 ms step. Next, we applied a Fast Fourier Transform to each segment to extract its frequencies. We squared the results and converted them to a logarithmic scale to better match how human ears perceive loudness. We visualized this data as a spectrogram: the horizontal axis is time, the vertical axis is frequency, and darker colors mean stronger sounds as the assignment ask. Finally, we calculated the total energy for each segment and drew it as a line over the spectrogram to show how the volume changes over time.

3 Pitch detection algorithm

To find the pitch of the speech, we used an autocorrelation method to look for repeating patterns in the sound. For each frame, we first checked the volume. If the frame was too quiet, we set the pitch to zero to ignore background noise. For the louder frames, we calculated the autocorrelation to find the strongest repeating signal. To make sure we only capture real human voices, we limited our search to a normal pitch range between 75 Hz and 500 Hz. Once we found the correct time delay for the pitch, we converted it into frequency. Finally, we plotted these pitch values as points on the spectrogram to show how the voice goes up and down.

4 Feature Identification

To identify the linguistic feature of a "Question Rise," which is characteristic of interrogative sentences, the algorithm analyzes the final pitch contour of the audio signal. This process involves extracting the last 20 frames that contain valid, non-zero pitch data. The algorithm then compares the fundamental frequency at the beginning of this segment to the frequency at its end. If the pitch exhibits an upward shift of more than 30 Hz across this window, the utterance is classified as having a rising intonation. When this feature is successfully detected, the corresponding time frames are visually emphasized with a transparent yellow highlight overlay on the final spectrogram. Finally, these results can be directly verified by examining the summary plot.

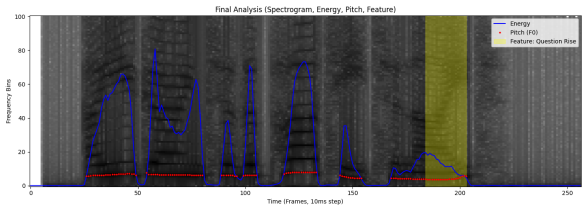


Figure 1: summary